

# Semistructured Data Integration (2)

TECHNOLOGIES FOR INFORMATION SYSTEMS

Letizia Tanca

Politecnico di Milano

# The new application context (recall)

- A (possibly large) number of data sources
- Heterogeneous data sources
- Different levels of data structure
  - Databases (relational, OO...)
  - Semi-structured data sources (XML, HTML, more markups ...)
  - Unstructured data (text, multimedia etc...)
- Different terminologies and different operational contexts
- Time-variant data (e.g. WEB)
- Mobile, transient data sources

# Ontology Formal Definitions

$$O = (C, R, I, A)$$

O: ontology, C: concepts, R: relations, A: axioms,  
I: Instances

- Often specified in some logic-based language
- Organized in a ISA hierarchy
- *I is an* instance collection, stored in the information source

## More Formal Definitions

An ontology is (part of) a knowledge base, composed by:

- a **T-Box**: contains the definitions of concepts and roles, and also contains all the axioms of our *logical theory* (e.g. “A father is a Man with a Child”).
- an **A-box**: contains all the basic assertions (also known as ground facts) of the logical theory (e.g. “Tom is a father” is represented as Father(Tom)). It describes *the instances and their relationships*.



# Example: OpenCyc

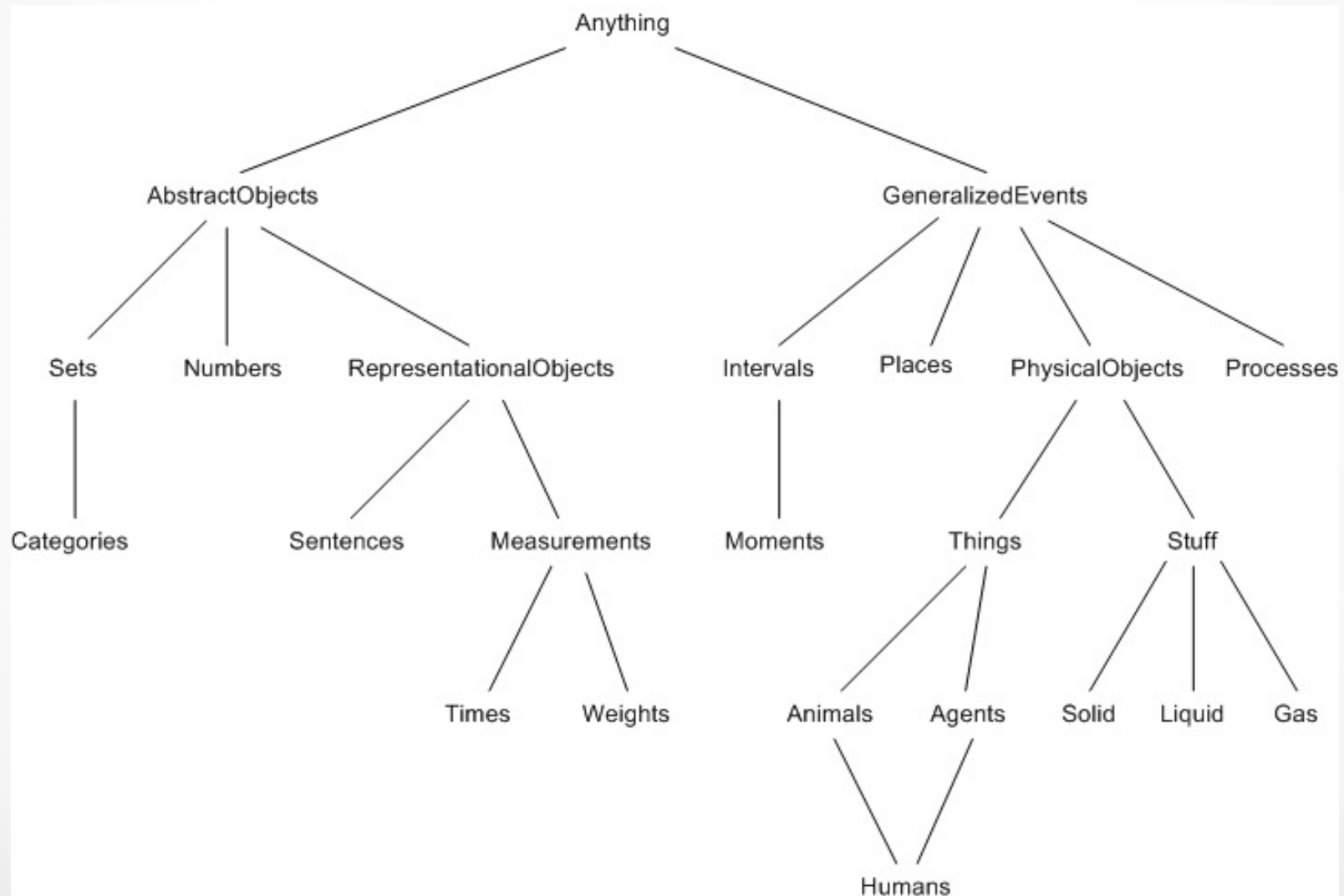
- The open-source version of the Cyc project was launched in July 1984 by Douglas Lenat at the Microelectronics and Computer Technology Corporation (MCC), a research consortium started by two United States-based corporations.
- Available until early 2017 as OpenCyc under an open source (Apache) license.
- Later, Cyc has been made available to AI researchers under a research-purpose license as ResearchCyc.
- The entire Cyc ontology contains hundreds of thousands of terms, along with millions of assertions relating the terms to each other, forming an ontology whose domain is all of human consensus reality.

●



# Another example:

## Top level concepts of the Russel and Norvig ontology



# Semantic Interoperability → *Semantic Web*

- A vision for the future of the Web in which information is given explicit meaning, making it easier for machines to **automatically process and integrate** information available on the Web.
- Built on **XML**'s ability to define customized tagging schemes and **RDF**'s flexible approach to representing data(\*).
- The first level above RDF: **OWL**, an ontology language that can formally describe the meaning of terminology used in Web documents → beyond the basic semantics of RDF Schema.

(\*) Also different implementations of RDF exist, not based on XML (e.g. Turtle)

# Linked Data

- **Linked Data** is a movement backed by **W3C (World Wide Web Consortium - <https://www.w3.org/>)** about connecting data sets across the Web. It describes a method of publishing semi-structured data so that it can be interlinked and become more useful.
- **W3C** builds upon standard Web technologies such as HTTP, RDF and URIs, but extends them to share information in a way that can be read automatically by computers, enabling data from different sources to be connected and queried.

# Linked Data (2)

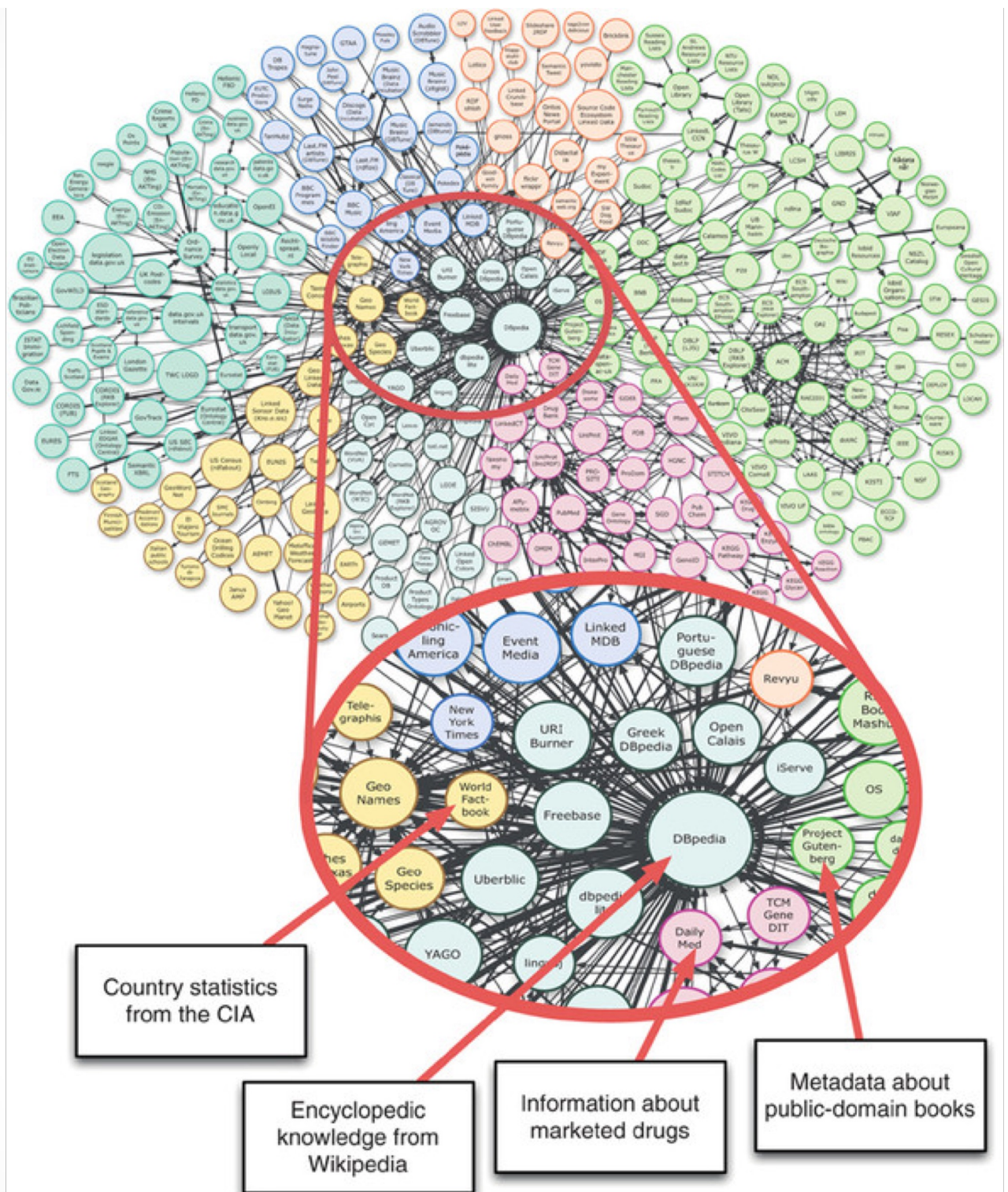
- Linked Data is a subset of the wider **Semantic Web** movement, which is about adding meaning to the Web (**Tim Berners-Lee**)

In general:

- Open Data describes data that have been uploaded to the Web and is accessible to all
- Linked Open Data extends the Web with a data commons by publishing various open datasets as RDF on the Web and by setting RDF links among them

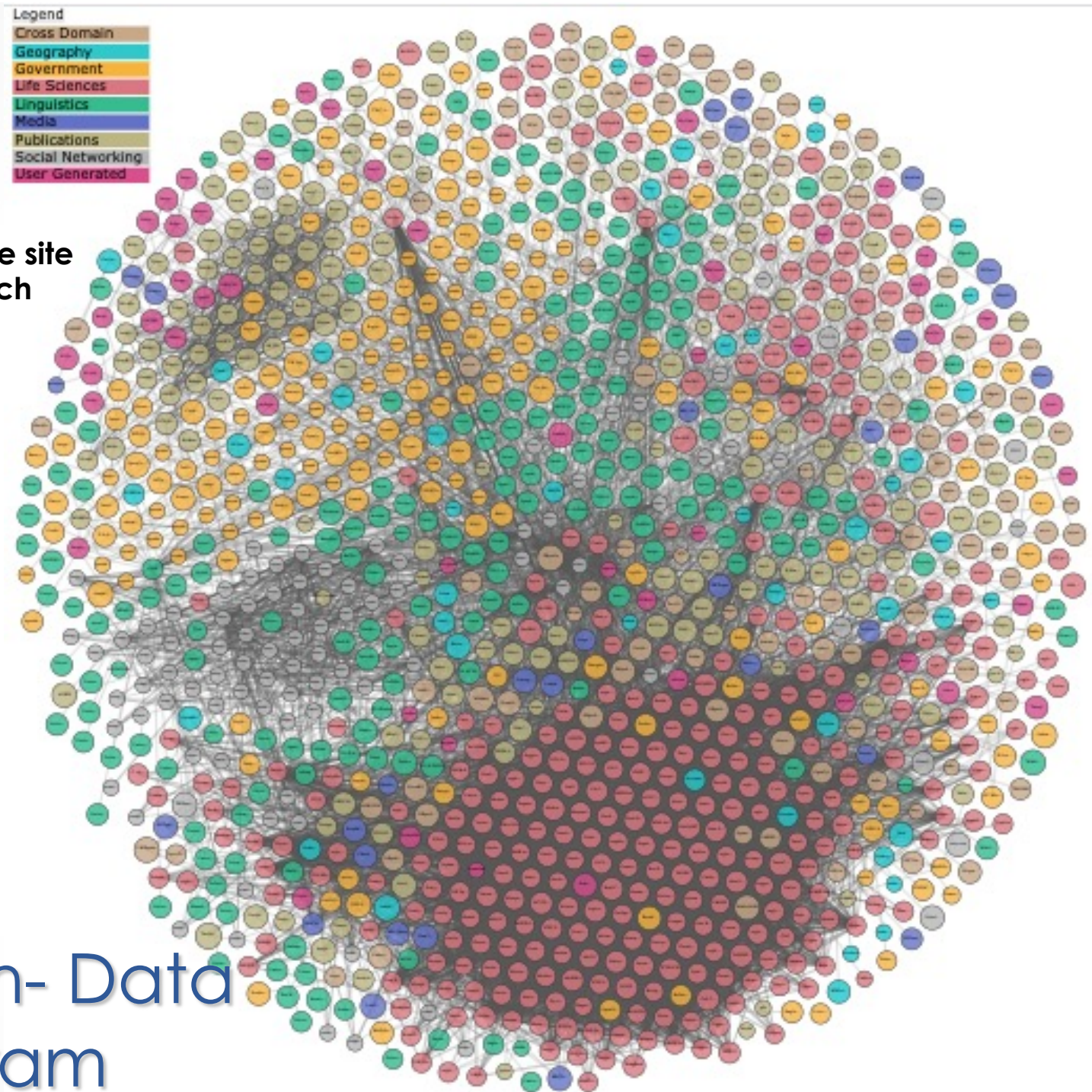


# Linked Open Data





These images come from the site <https://lod-cloud.net/>, which is actually updated very frequently (last update Sept 2 2025). Visit the site: it's interesting and contains more info!



# Linked-Open-Data Cloud Diagram



## Subclouds by domain

Cross-Domain

Geography

Government

Life Sciences

Linguistics

Media

Publications

Social Networking

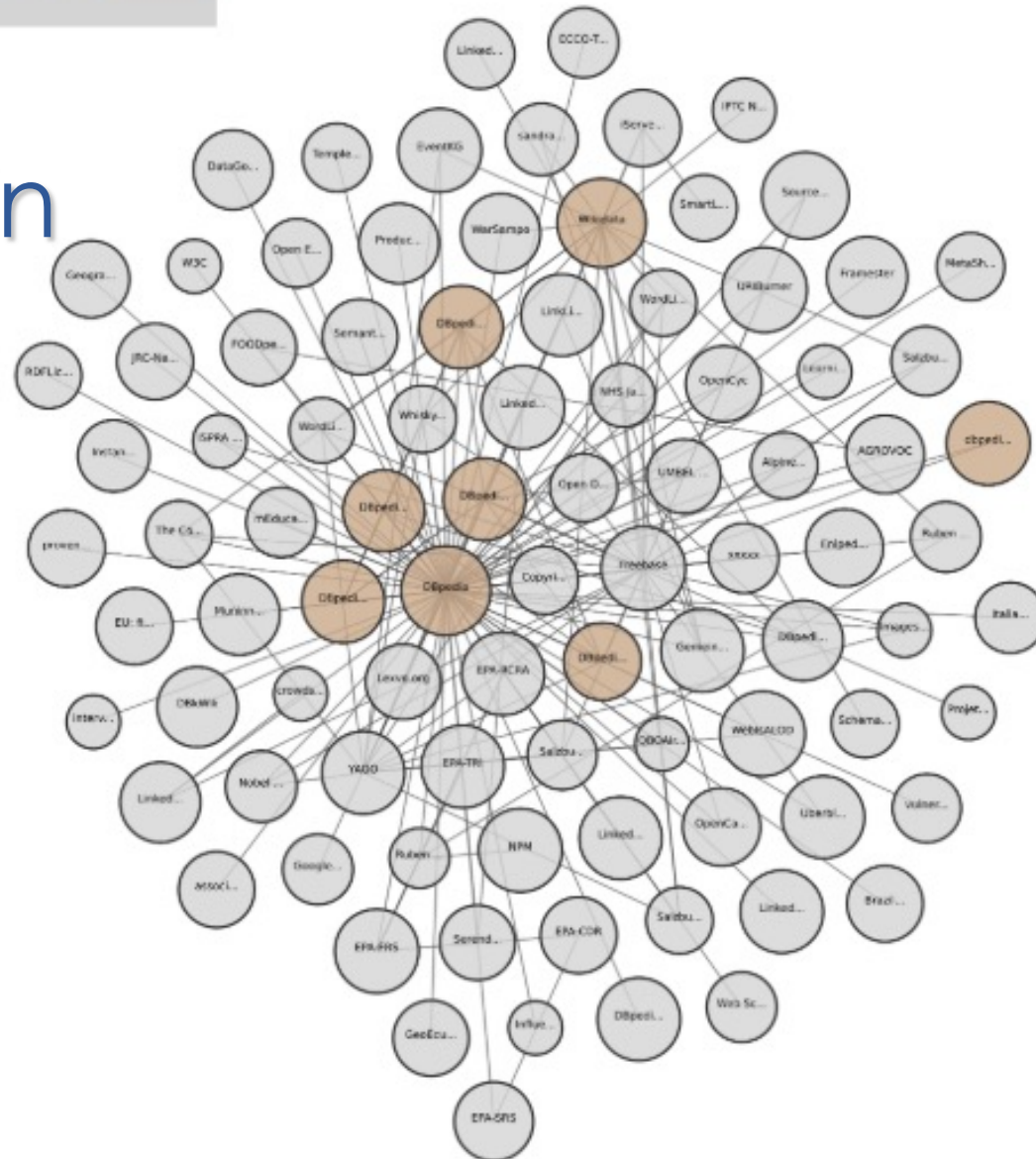
User-Generated

### Legend

Wikipedia Related

Other

# Cross-Domain Subcloud



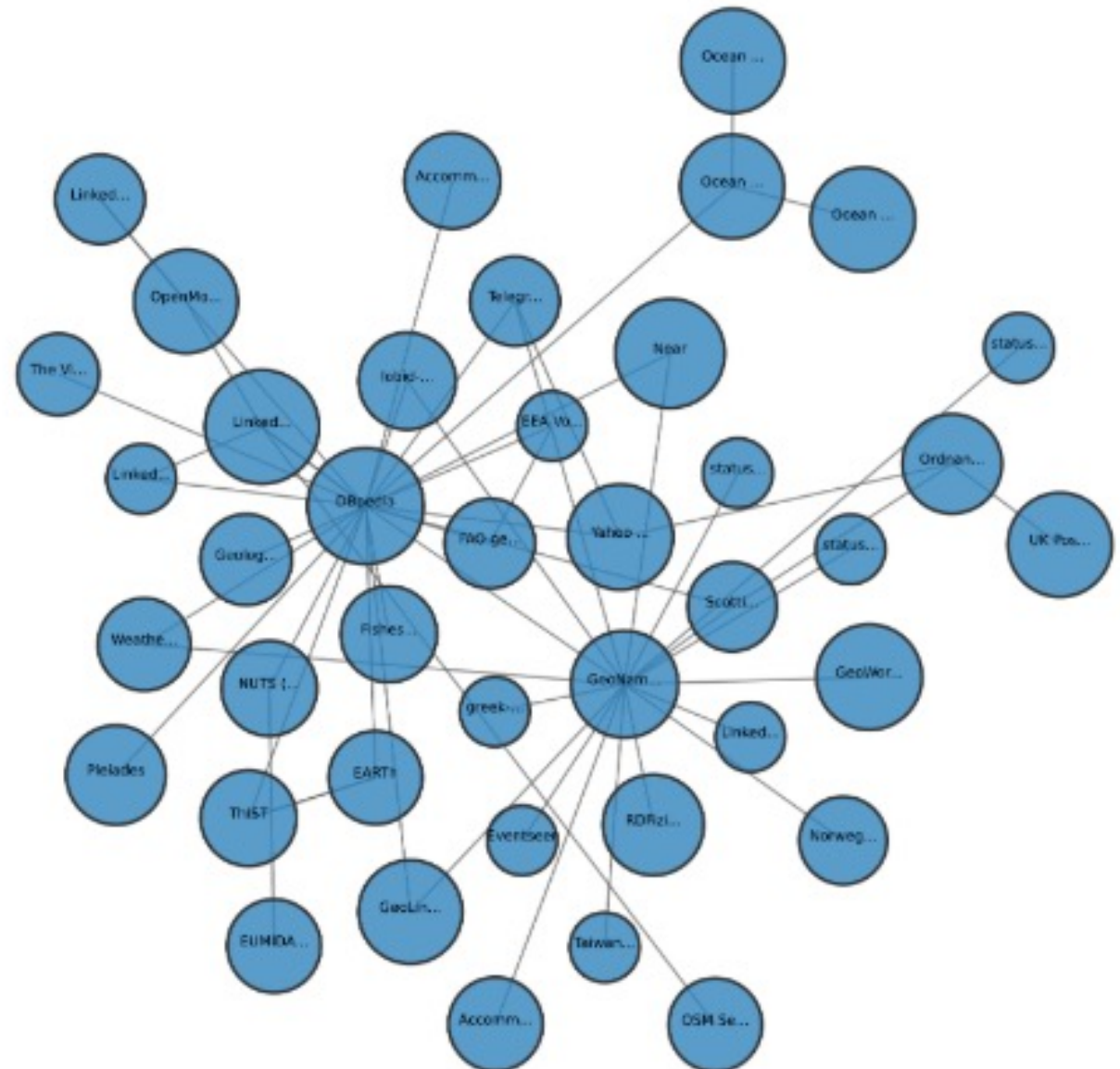
The Cross-Domain Linked Open Data Cloud from lod-cloud.net



Legend

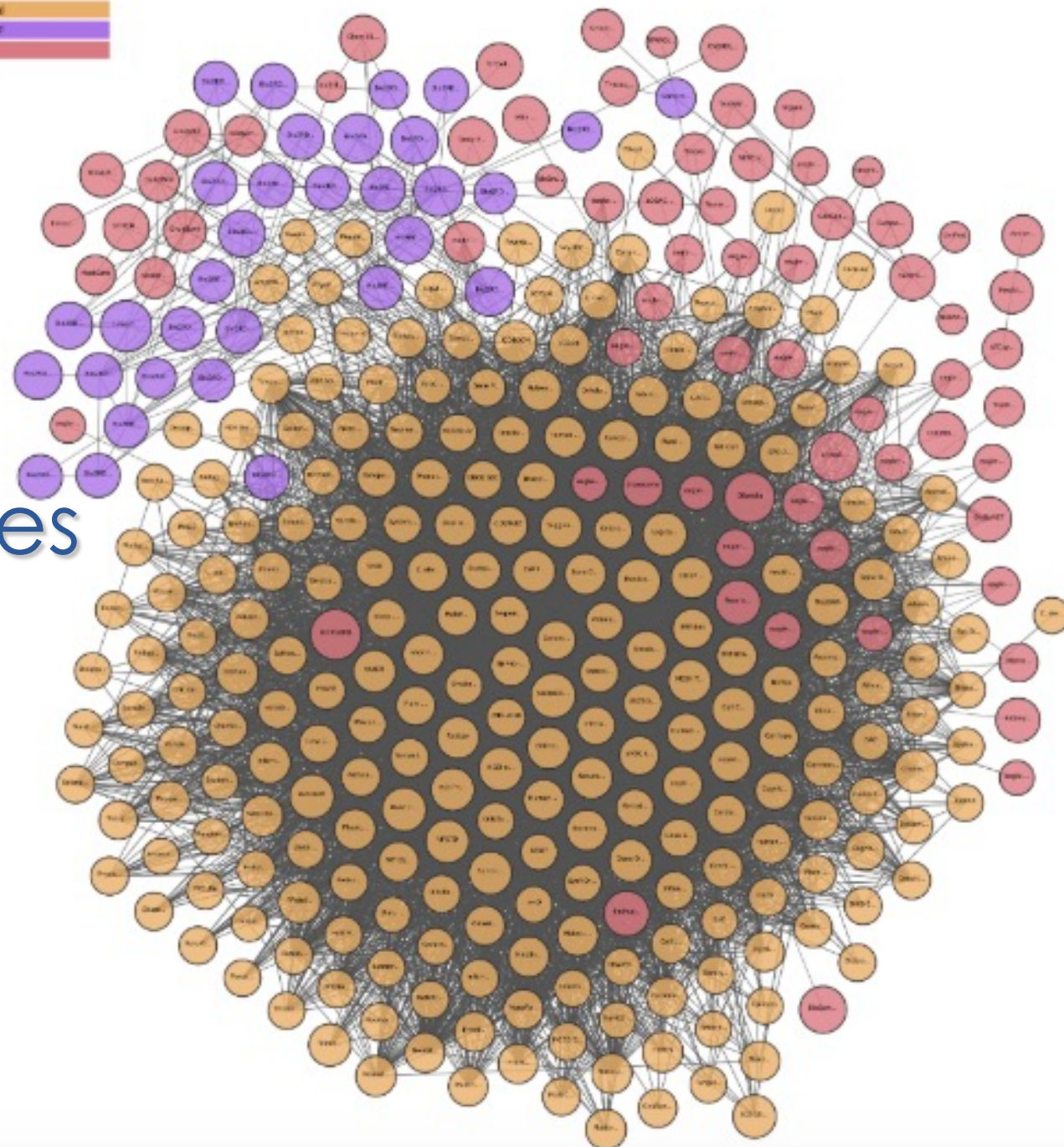
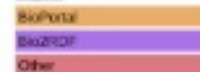
## Geography

# Geography Subcloud





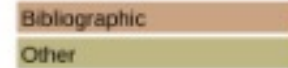
Legend



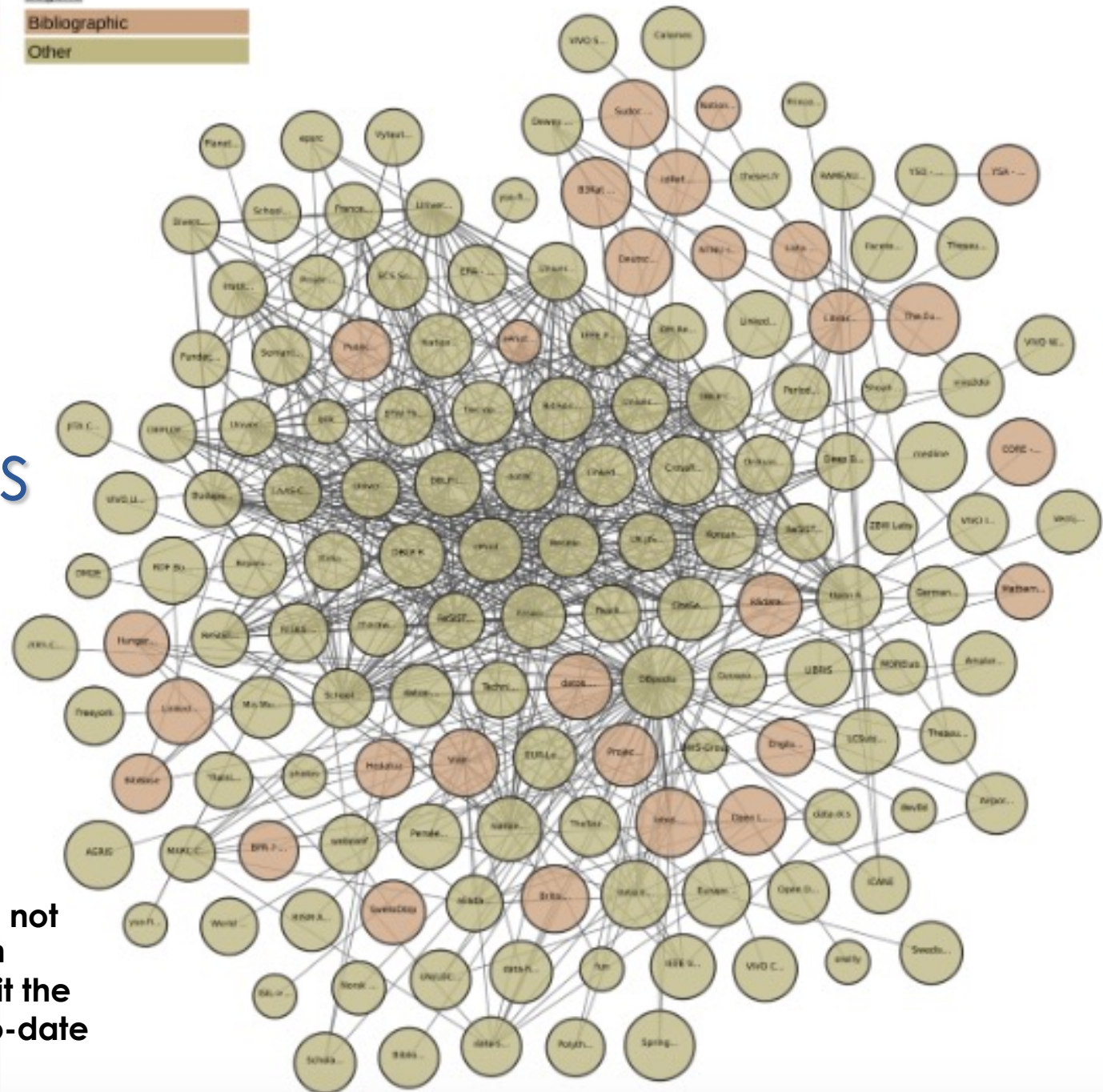
# Life Sciences Subcloud



Legend



# Publications Subcloud



Caution: these images are not the most recent ones from <https://lod-cloud.net/> : visit the site if you want to be up-to-date

# Some famous datasets (1)

- CKAN – registry of open data and content packages provided by the Open Knowledge Foundation - <https://ckan.org/>
- DBpedia – a dataset containing data extracted from Wikipedia : <https://www.dbpedia.org/>
  - DBpedia is a crowd-sourced community effort to extract structured content from the information created in various Wikimedia projects. This structured information resembles an open knowledge graph (OKG) which is available for everyone on the Web.
  - A knowledge graph is a special kind of database which stores knowledge in a machine-readable form and provides a means for information to be collected, organised, shared, searched and utilised. it contains about 5 million concepts described by some billion triples, including abstracts in more than ten different languages.
  - Recent info: <https://www.dbpedia.org/blog/>



# Some famous datasets (2)

- GeoNames provides RDF descriptions of more than 8,000,000 geographical features worldwide - <https://www.geonames.org/>.
- YAGO (Yet Another Great Ontology) is an ever-growing open-source knowledge base developed at the Max Planck Institute for Computer Science in Saarbrücken in collaboration with the DIG group at Télécom ParisTech University.. It is automatically extracted from Wikipedia and GeoNames. - <https://www.mpi-inf.mpg.de/departments/databases-and-information-systems/research/yago-naga/yago:>
- Currently, YAGO has knowledge of more than 10 million entities and contains more than 120 million facts about these entities.
- UMBEL – a lightweight reference structure that has links to 1.5 million named entities from DBpedia and YAGO
- FOAF (Friend of a Friend) - is a machine-readable ontology used to describe people, their activities, and their relationships on the web.

•

•

# The RDF language

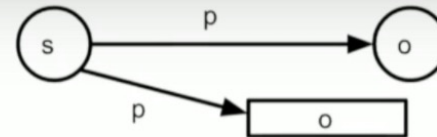
At the core of RDF is this notion of a triple **subject-predicate-object**, a statement that represents two vertices connected by an edge:

- **Subject:** a resource, or a node in the graph
- **Predicate:** an edge – a relationship
- **Object:** another node or a literal value



## RDF

(s p o)



### Vertices

Resources : URIs

Attribute Values : Literal Values

### Edges

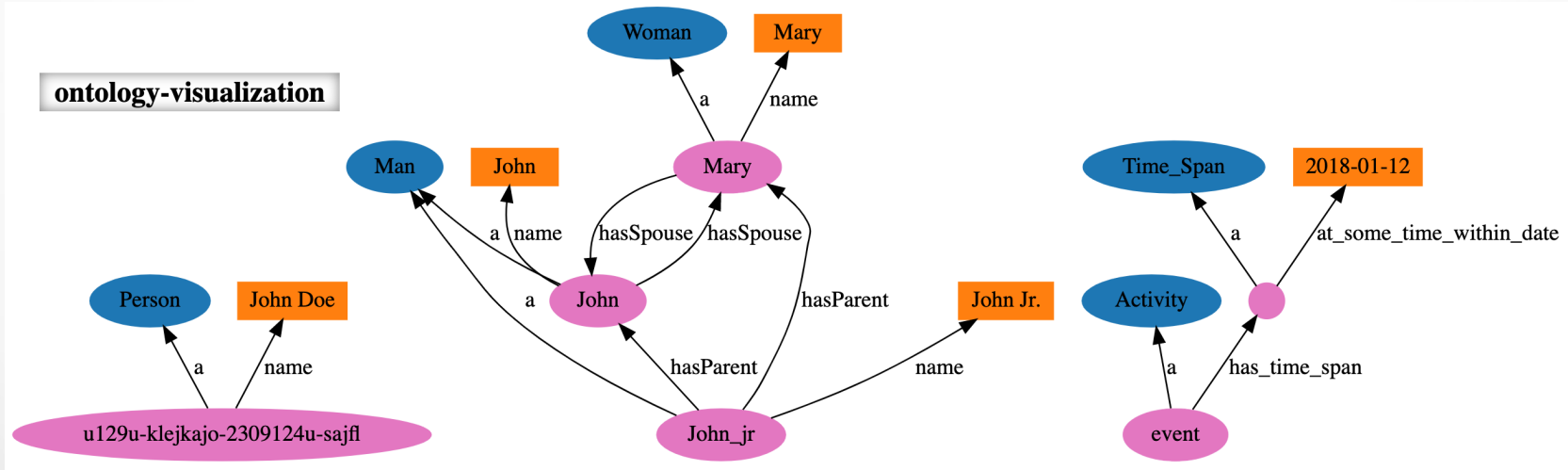
Relationships : URIs

Nodes or Edges have NO internal structure

# A sample RDF file

```
:John a :Man ;  
      :name "John" ;  
      :hasSpouse :Mary .  
:Mary a :Woman ;  
      :name "Mary" ;  
      :hasSpouse :John .  
:John_jr a :Man ;  
      :name "John Jr." ;  
      :hasParent :John, :Mary .  
:Time_Span a owl:Class .  
:event a :Activity ;  
      :has_time_span [  
        a :Time_Span ;  
        :at_some_time_within_date "2018-01-12"^^xsd:date  
      ] .  
:u129u-klejkajo-2309124u-sajfl a :Person ;  
      :name "John Doe" .
```

## ontology-visualization



A fragment of an **RDF**  
(XML) document,  
describing an  
ontology.

The language is **OWL**  
[http://www.w3.org/TR  
/owl-ref/](http://www.w3.org/TR/owl-ref/)

```
<?xml version="1.0"?>
  <rdf:RDF
    xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
    xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
    xmlns:owl="http://www.w3.org/2002/07/owl#"
    xmlns:daml="http://www.daml.org/2001/03/daml+oil#"
    xmlns="http://eng.it/ontology/tourism#"
    xmlns:dc="http://purl.org/dc/elements/1.1/"
    xml:base="http://eng.it/ontology/tourism">
    <owl:Ontology rdf:about=""/>
    <owl:Class rdf:ID="Church">
      <rdfs:comment rdf:datatype="http://www.w3.org/2001/XMLSchema#string"
        >Definition: Edificio sacro in cui si svolgono pubblicamente gli atti
        di culto delle religioni cristiane.</rdfs:comment>
      <rdfs:subClassOf>
        <owl:Class rdf:about="#PlaceOfWorship"/>
      </rdfs:subClassOf>
    </owl:Class>
    <owl:Class rdf:ID="Theatre">
      <rdfs:comment rdf:datatype="http://www.w3.org/2001/XMLSchema#string"
        >Definition: a building where theatrical performances or motion-
        picture shows can be presented.</rdfs:comment>
      <rdfs:subClassOf>
        <owl:Class rdf:about="#SocialAttraction"/>
      </rdfs:subClassOf>
    </owl:Class>
    <owl:Class rdf:ID="DailyCityTransportationTicket">
      <rdfs:subClassOf>
        <owl:Class rdf:about="#CityTransportationTicket"/>
      </rdfs:subClassOf>
    </owl:Class>
    <rdfs:comment rdf:datatype="http://www.w3.org/2001/XMLSchema#string"
      >Definition: Biglietto che consente di usufruire di un numero
      illimitato di viaggi sui mezzi pubblici (autobus e metropolitana)
      all'interno del centro urbano (o della regione, con un costo maggiore)
      per un periodo di 24 ore.</rdfs:comment>
    </owl:Class>
```



# OWL

- Designed to meet the need for a Web Ontology Language, **OWL** is part of the growing stack of **W3C recommendations** related to the Semantic Web.
- The OWL Web Ontology Language is designed **for use by applications that need to process the content of information** instead of just *presenting* information to humans.
- OWL facilitates greater **machine interpretability** of Web content than that supported by XML, RDF, and RDF Schema (RDF-S) by **providing additional vocabulary along with a formal semantics**.
- OWL has three increasingly-expressive sublanguages: **OWL Lite**, **OWL DL**, and **OWL Full**.

# XML, RDF and OWL

- **XML** provides a surface syntax for structured documents but **imposes no semantic constraints** on the meaning of these documents.
- **XML Schema** is a language for restricting the structure of XML documents and also extends XML with data types.
- **RDF** is a data model for objects ("resources") and relations between them, provides a simple semantics for this data model, and can be represented in an XML syntax.
- **RDF Schema** is a vocabulary for describing properties and classes of RDF resources, with a semantics for generalization-hierarchies of such properties and classes.
- **OWL** adds more vocabulary for describing properties and classes: among others, relations between classes (e.g. disjointness), cardinality (e.g. "exactly one"), equality, richer typing of properties, characteristics of properties (e.g. symmetry), and enumerated classes.

# Reasoning services for ontologies

## Services for the Tbox

- **Subsumption:** verifies if a concept C subsumes (is a subconcept of) another concept D
- **Consistency:** verifies that there exists at least one interpretation I which satisfies the given Tbox
- **Local Satisfiability:** verifies, for a given concept C, that there exists at least one interpretation in which C is true.

## Services for the Abox

- **Consistency:** verifies that an Abox is consistent with respect to a given Tbox
- **Instance Checking:** verifies if a given individual x belongs to a particular concept C
- **Instance Retrieval:** returns the extension of a given concept C, that is, the set of individuals belonging to C.

# Comparison

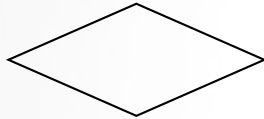
- Analysis of the features of a **descriptive ontology** (data structures, instance management, constraint definition, queries)
- Comparison of these features with the functionality provided by *the usual representation approaches from the database world*



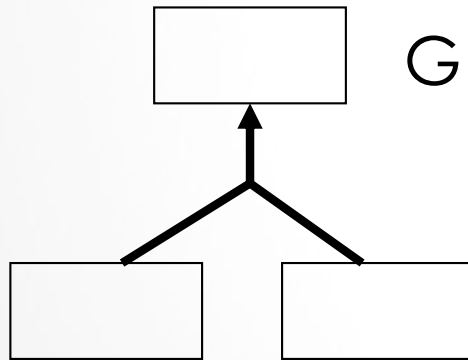
# ER vs.ontology



ENTITY



RELATIONSHIP



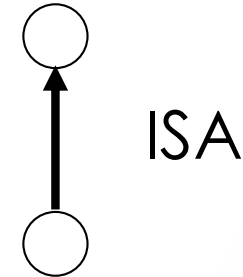
GENERALIZATION  
HIERARCHY



ATTRIBUTE

○ CONCEPT

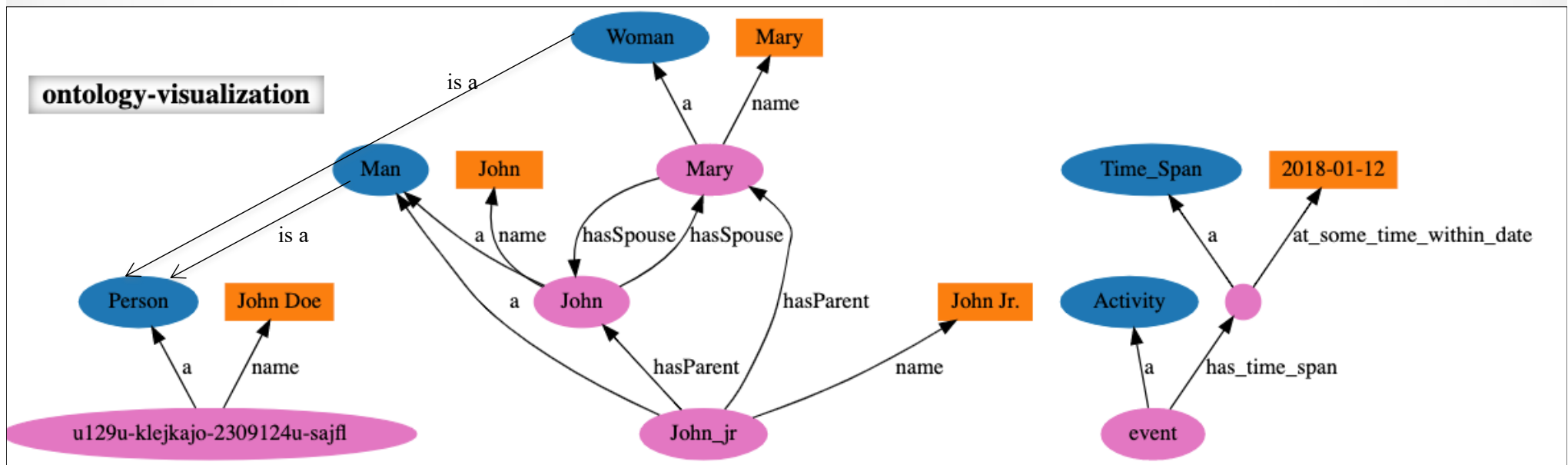
RELATION



ISA

DATA TYPE PROPERTY

# Let's see how this corresponds to an ER diagram



- **Caution:** An ER schema does not have VALUES !

# Comparison

Descriptive ontologies require rich models to enable representations close to human perception

|                                           | <b>Ont.</b> | <b>DB</b>  |
|-------------------------------------------|-------------|------------|
| Complex data structures                   | <b>No</b>   | <b>yes</b> |
| Generalization/specialization hierarchies | <b>yes</b>  | <b>yes</b> |
| Defined concepts                          | <b>yes</b>  | <b>no</b>  |

# DBs versus ontologies

How could we improve database conceptual models to fulfill ontology requirements ?

- Supporting defined concepts and adding the necessary reasoning mechanisms
- Managing missing and incomplete information: semantic differences between the two assumptions made w.r.t. missing information (*Closed World Assumption* vs. *Open World Assumption*)
- Databases are assumed to represent certain data: a tuple in the database is true, any tuple NOT in the database is false (*Closed World Assumption*)

# Ontologies as a support for integration problems

- **Discovery** of “equivalent” concepts (**mapping**)
  - What does **equivalent** mean? → again we look for some kind of **similarity**
- **Formal representation** of these mappings
  - How are these mappings represented?
- **Reasoning** on these mappings
  - How do we use the mappings within our reasoning and query-answering process?



# Ontology matching

- The process of finding pairs of resources coming from different ontologies which can be considered *equal in meaning* – *matching operators*
- Again we need some kind of *similarity measure*.
- Recall: a *similarity value* is usually a number in the interval  $[0,1]$
- *Caution:* this time the similarity measure takes into account *semantics*, not only the structure of the words as in the examples given in the previous lectures !!!!  
•

# More on similarity

- The concept of similarity is a basic concept in human cognition.
- Similarity plays an essential role in taxonomy, recognition, case-based reasoning and many other fields. There are many aspects of the concept of similarity that have eluded formalization.
- According to Zadeh(\*), "Formulation of a valid, general-purpose definition of similarity is a challenging problem".
- There do exist many special-purpose definitions which have been employed with success in cluster analysis, search, classification, recognition and diagnostics.
- (\*) The "inventor" of *fuzzy sets* and *fuzzy logic*

# As already seen, similarity is strictly related to distance

*Self-identity* is the property which says that the distance between identical objects is zero. This translates to the following self-identity axiom:

**Axiom 2.1.1.** *For all  $x$  in  $S$ ,  $d(x, x) = 0$ .*

*Positivity* is the property which says that distinct objects have a nonzero distance:

**Axiom 2.1.2.** *For all  $x \neq y$  in  $S$ ,  $d(x, y) > 0$ .*

*Symmetry* says that the order of two elements does not matter for the distance between them:

**Axiom 2.1.3.** *For all  $x$  and  $y$  in  $S$ ,  $d(x, y) = d(y, x)$ .*

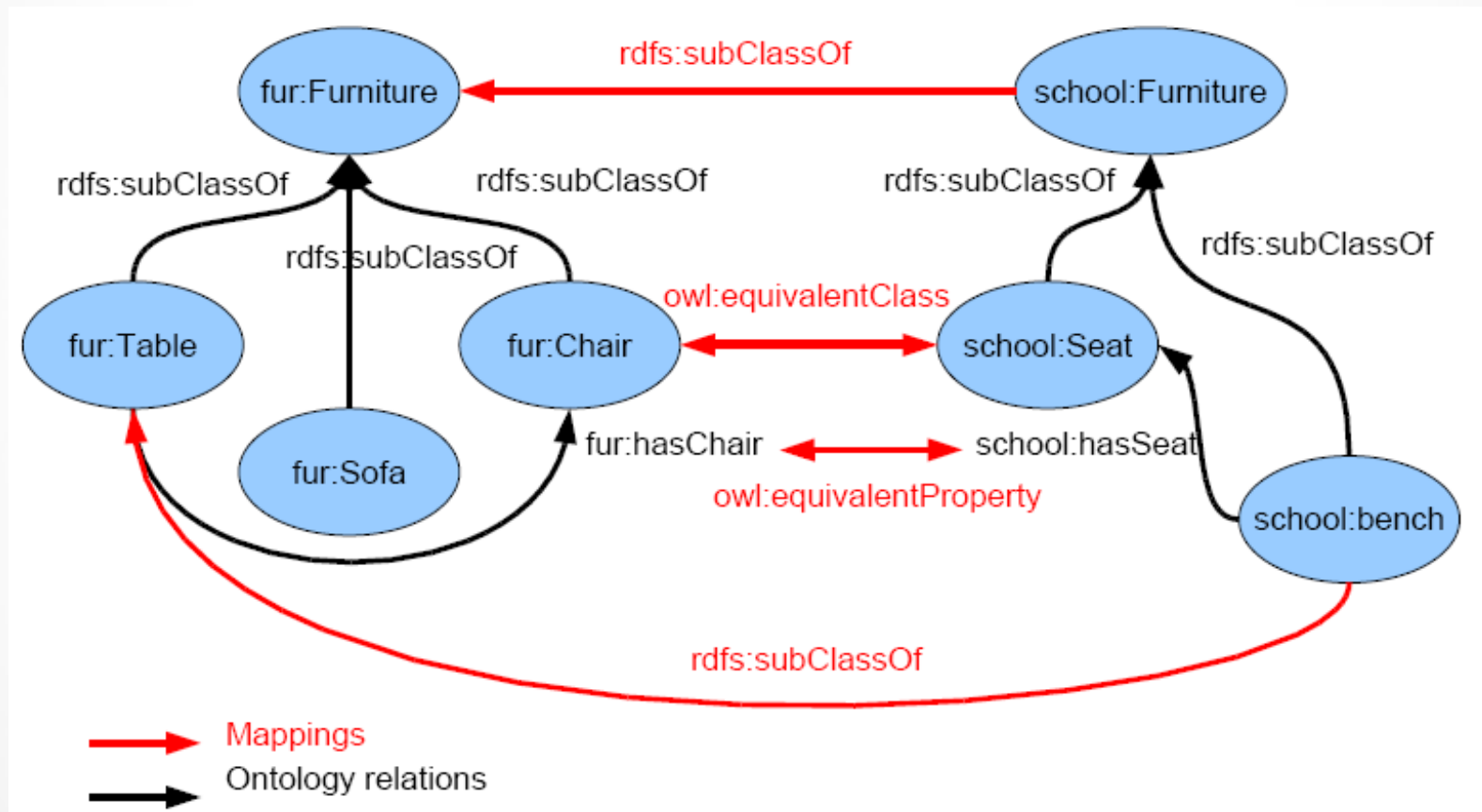
The *triangle inequality* says that the distance between  $y$  and  $z$  does not exceed the sum of the distance between  $y$  and  $x$  and the distance between  $x$  and  $z$ :

**Axiom 2.1.4.** *For all  $x, y, z \in S$ ,  $d(y, z) \leq d(y, x) + d(x, z)$ .*

# Ontology *mapping*

- The process of relating similar concepts or relations of two or more information sources using equivalence relations or order relations.
- These relations are commonly implemented in inference and reasoning softwares, so we can use the output ontology to perform complex tasks on them without extra effort.

# Ontology mapping





# Reasons for ontology mismatches

At the **definition language** level:

- Syntax
- Availability of different constructs (e.g. part-of, synonym, etc.)
- Linguistic primitives' semantics (e.g. union or intersection of multiple intervals)

→ Normalize by translating to the same language/  
paradigm

# Reasons for ontology mismatches

At the **ontology** level:

- **Scope:** Two classes seem to represent the same concept, but do not have exactly the same instances
- **Model coverage and granularity:** a mismatch in the part of the domain that is covered by the ontology, or the level of detail to which that domain is modelled.
- **Paradigm:** Different paradigms can be used to represent concepts such as time. For example, one model might use temporal representations based on continuous intervals while another might use a representation based on discrete sets of time points.
- **Encoding**
- **Concept description:** e.g. a distinction between two classes can be modeled using a qualifying attribute or by introducing a separate class, or the way in which is-a hierarchy is built
- **Homonymies**
- **Synonymies**

# Recall the steps of Data Integration



## Schema Reconciliation

Schema reconciliation:  
mapping the **data structure**

## Record Linkage

Record linkage: data  
matching based on **the  
same content**

## Data Fusion

Data fusion: reconciliation  
of **non-identical content**

# How can ontologies support integration ?

An ontology as a **schema integration support** tool

- Ontologies used to represent the semantics of schema elements (if the schema exists)
- Similarities between the source ontologies guide conflict resolution
  - At the schema level (if the schemata exist)
  - At the instance level (record linkage)

An ontology **instead of a global schema:**

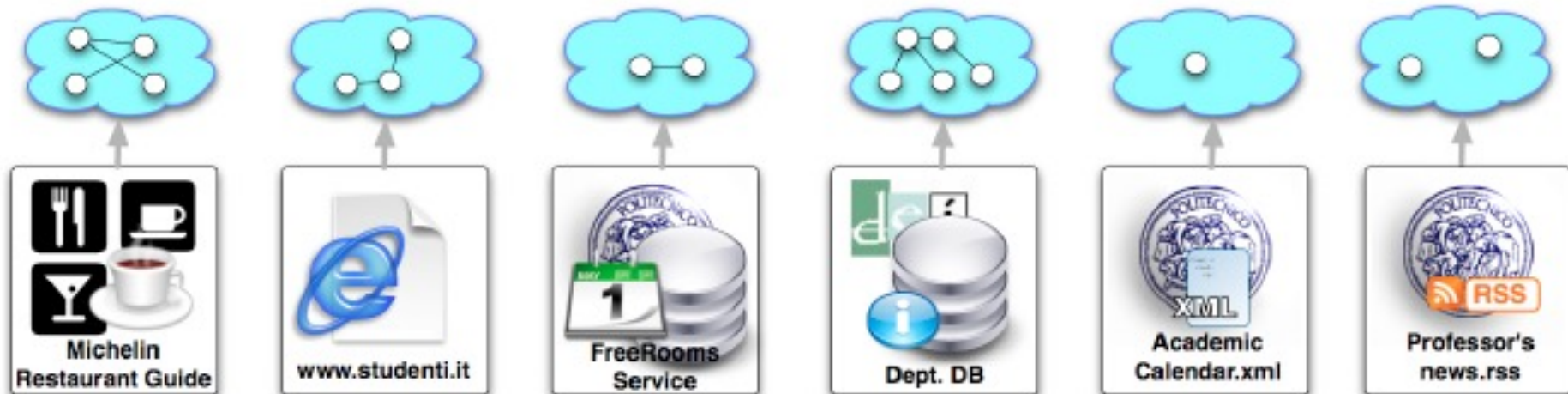
- Schema-level representation only in terms of ontologies
- Ontology mapping, merging, etc. instead of schema integration
- Integrated ontology used as a schema for querying

# An ontology instead of a global schema

## Recall the students' example we made for Mediation

- Data-source heterogeneity is solved by extracting the semantics in an ontological format (potentially *at run-time*)
- Automatic Wrapper generation + Query translation will bridge among two models.
- Not an easy task:
  - several issues, e.g., impedance mismatch
  - unstructured data sources

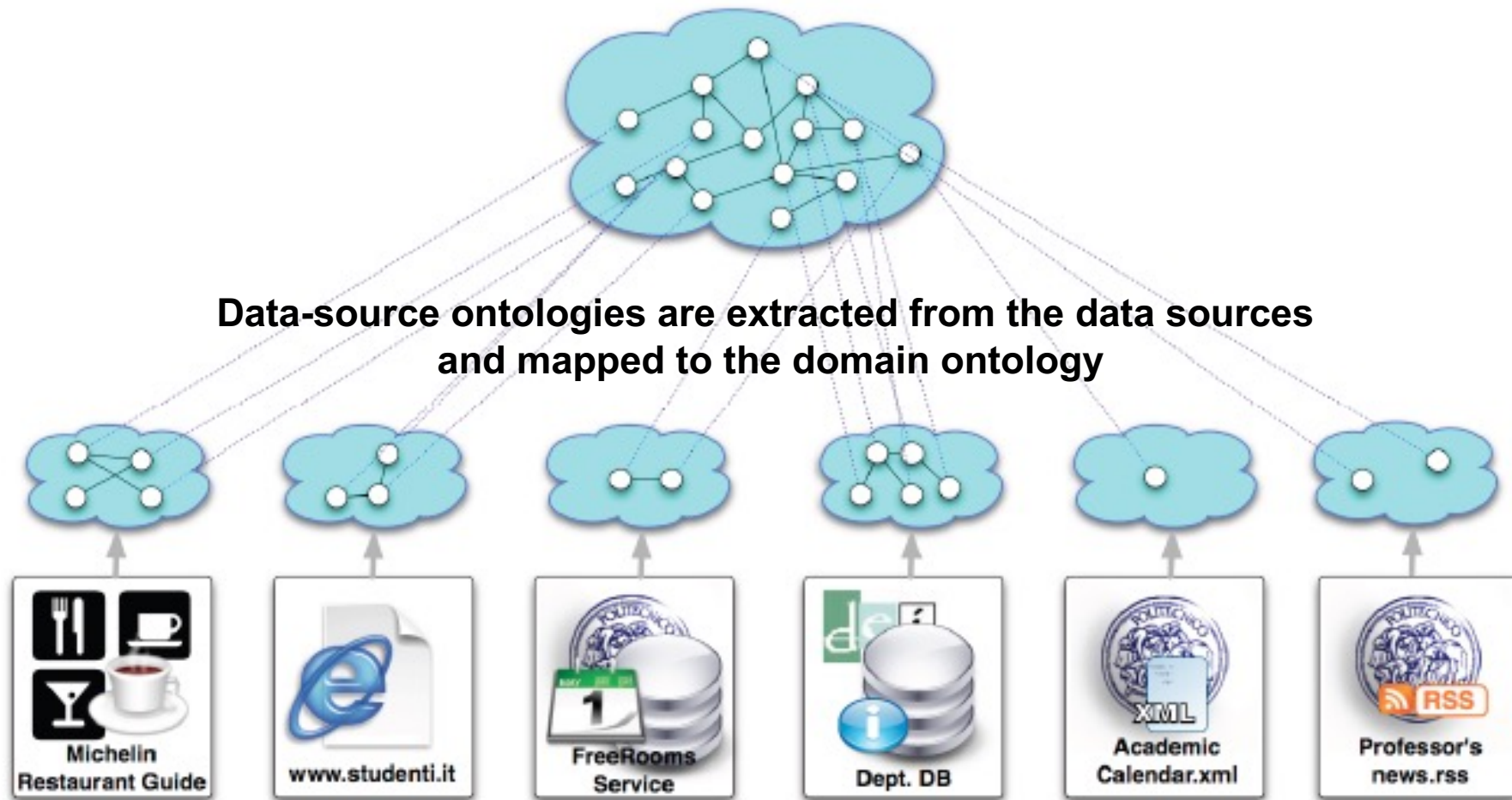
**Note:** the impedance mismatch is the problem that occurs when the database model is different from the programming language model (and vice-versa), e.g. OO and Relational





# An ontology instead of a global schema

Global Schema : Domain Ontology (*at design-time*)



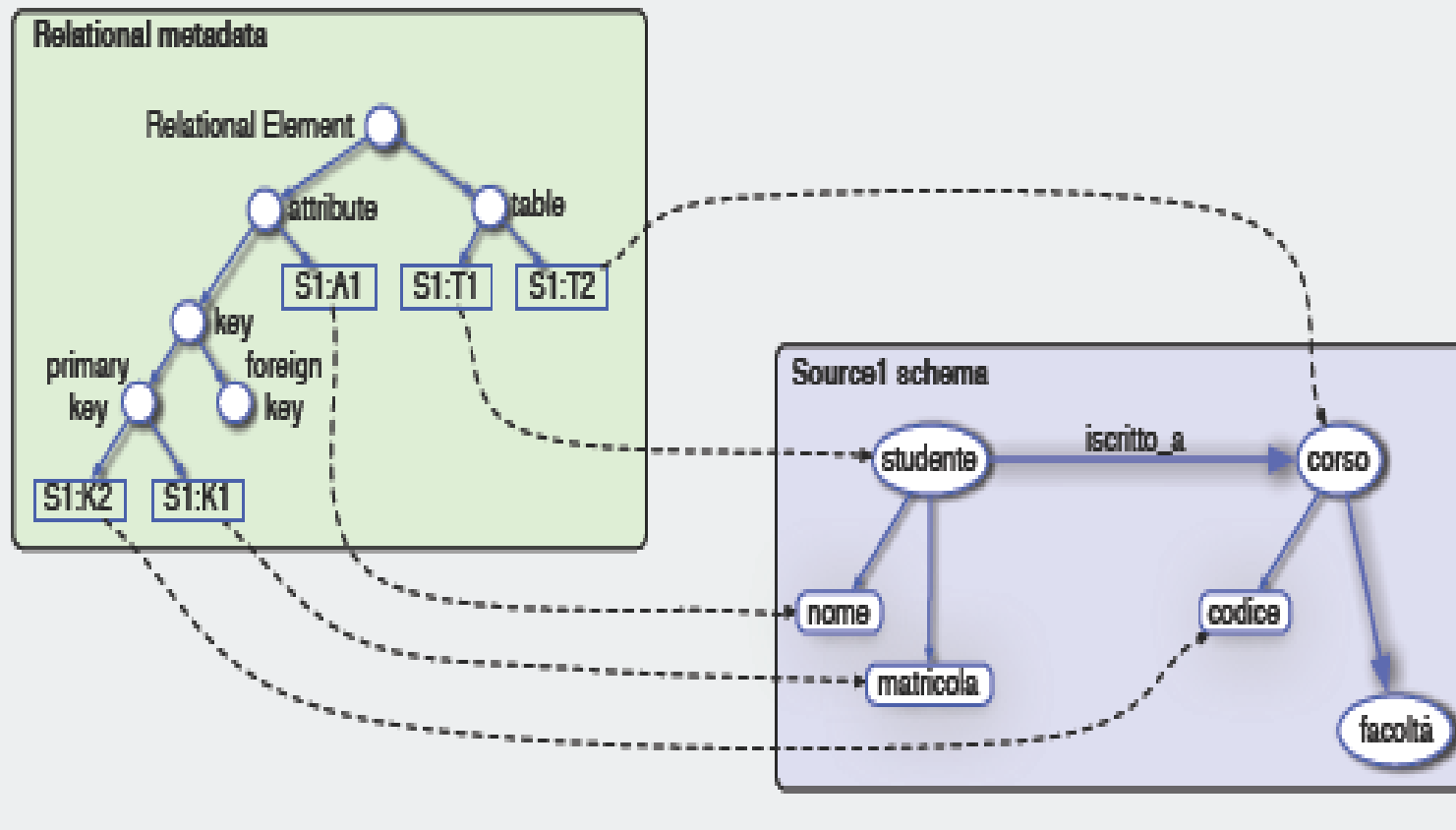
# More:

## how can ontologies support integration?

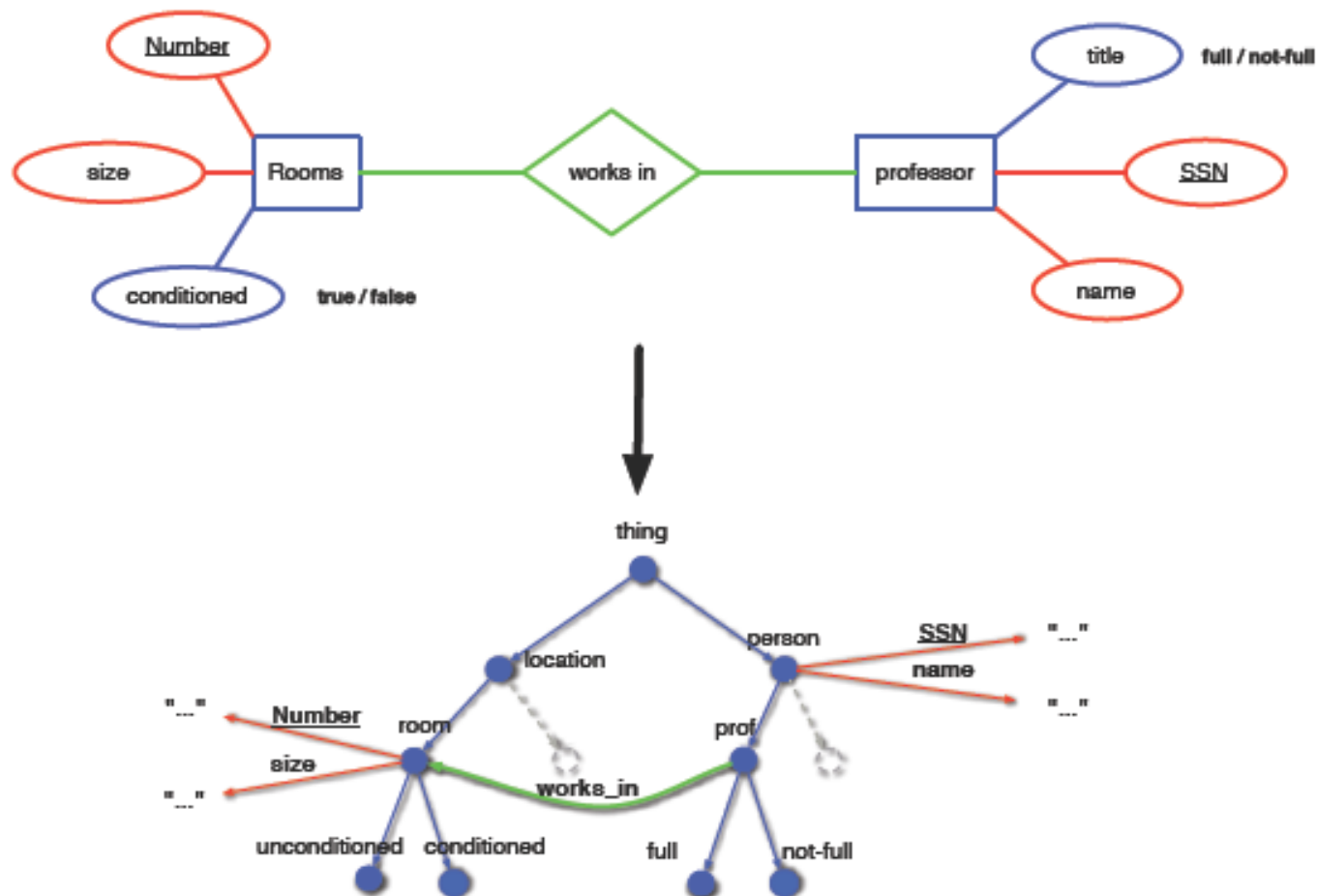
- An ontology as a support tool for content interpretation and wrapping (e.g. HTML pages)
- An ontology as a mediation support tool for content inconsistency detection and resolution (record linkage and data fusion)

# Ontology extraction from a relational schema: example

## Source extracted SCHEMA



# Ontology extraction from a ER schema



# Ontology query processing

Ontologies require **query languages** as well, for:

- Schema exploration (when the schema is replaced by an ontology)
- Reasoning on the schema
- Instance querying (when the instance is contained in an ontology, like in the Semantic Web case)
- Example of ontology query language: SPARQL (W3C)

# Examples of SPARQL queries

We can give an informal idea of **SPARQL** focusing on the SELECT queries. In particular we can say that a query  $q$  is a structure like:

*SELECT ?X1 ... ?Xn WHERE P*

With  $?X1 \dots ?Xn$  as the set of variables and  $P$  as the graph pattern.

Let's consider this **example** where we have a simple SPARQL query which asks for all projects in which some PhD student is involved:

*SELECT ?Y*

*WHERE { ?X rdf:type PhDStudent. ?X inProject ?Y }*

This next **example** of query instead retrieves employees who are PhD students or professors together with their projects:

*SELECT ?X ?Y*

*WHERE { { ?X rdf:type PhDStudent. UNION ?X rdf:type Professor. }  
AND ?X inProject ?Y. }*



# Ontology query processing versus database query processing

When we use ontologies to interact with databases, we have to take care of:

- The transformation of the ontological query into the language of the datasource, and the other way round
- IMPORTANT : the **different semantics** → **CWA** (Closed World Assumption) versus **OWA** (Open World Assumption)!!!!
- What has to be processed where (e.g. push the relational operators to the relational engine)

# The new application context (recall)

- A (possibly large) number of data sources
- Heterogeneous data sources
- Different levels of data structure
  - Databases (relational, OO...)
  - Semi-structured data sources (XML, HTML, more markups ...)
  - Unstructured data (text, multimedia etc...)
- Different terminologies and different operational contexts
  - Time-variant data (e.g. WEB and social media)
  - Mobile, transient data sources (e.g. sensor values)

.....as you can see, everything becomes more and more dynamic. This course does not delve into this problem, which is, anyway, to be taken into account

# Next Lectures

From next lecture, we'll talk about the second important subject of this course:

**Data Warehouses:**  
a fully consolidated paradigm for *business analytics*.

After that, we'll come back to the frontiers of Data Management and Integration, which at the moment still constitute hot research topics

# Bibliography

- A. Doan, A. Halevy and Z. Ives, Principles of Data Integration, Morgan Kaufmann, 2012
- L. Dong, D. Srivastava, Big Data Integration, Morgan & Claypool Publishers, 2015
- Roberto De Virgilio, Fausto Giunchiglia, Letizia Tanca (Eds.): Semantic Web Information Management – A Model-Based Perspective. Springer 2009, ISBN 978-3-642-04328-4
- M. Lenzerini, Data Integration: A Theoretical Perspective, Proceedings of ACM PODS, pp. 233-246, ACM, 2002, ISBN: 1-58113-507-6
- Clement T. Yu, Weiyi Meng, Principles of Database Query Processing for Advanced Applications , Morgan Kaufmann, 1998, ISBN: 1558604340